MIL-M-36431B
27 January 1970
SUPERSEDING
MIL-M-36431A (DM)
13 April 1966 and
MIL-M-36431 (DM)
4 June 1965 and
MIL-M-36168 (DM)
8 November 1963

MILITARY SPECIFICATION

MASK, SURGICAL, DISPOSABLE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers disposable lightweight surgical masks, suitable for use by medical personnel in operating rooms and other areas where it is necessary to maintain protective asceptic techniques.

1.2 Classification. The masks shall be of the following types, as specified in the invitation for bids.

Type I — Mask, Surgical, Nonwoven Fabric, Blue, Disposable.

Type II — Mask, Surgical, Fibrous Glass or Polypropylene and Nonwoven Fabric, Blue or Green, Disposable.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Box, Folding, Paperboard.
PPP-B-585	Box, Wood, Wirebound.
PPP-B-601	Box, Wood, Cleated-Plywood.
PPP-B-621	Box, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.

FSC 6515

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Military

MIL-L-10547 Liners, Case, and Sheet, Overwrap, Water-vaporproof or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

JOURNAL OF BACTERIOLOGY

Volume 83 Journal of Bacteriology

(Application for copies should be addressed to the Williams and Wilkins Company, Baltimore Md. 21202.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

Munsell Color Company Color Samples (glossy finished).

(Application for samples should be addressed to Munsell Color Company, Inc., 2441 North Calvert Street, Baltimore, Md. 21218.)

3. REQUIREMENTS

3.1 Material. Except as specified herein, material shall conform to the applicable portions of the specifications and standards listed, and shall be capable of meeting the requirements shown therein. When materials are not specifically designated, those used shall be of the highest commercial quality, and entirely suitable for the purpose intended.

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3.1.1 Type I Mask.

3.1.1.1 Fabric. The body of the mask shall be a seamless, porous, nonwoven fabric consisting of a mixture of polyester and rayon fibers, unified with a water-resistant resin. Testing shall be as specified in 4.5.

3.1.1.1.1 Color. The color of the body of the mask shall be blue approximating the Munsell Color Company's Color Sample Code 10B 6/10, glossy finished, except that the color need not be uniform in the ribbing and periphery areas. The inside area of the body of the mask shall be light blue approximating the Munsell Color Company's Color Sample Code 10B 8/4, except that the color need not be uniform throughout.

3.1.1.1.2 Porosity of mask fabric. The porosity shall be such that (a) the pressure differential across a sample shall not exceed 0.09 inches of water when a 17.5 square inch specimen from the center of the mask is exposed to an enclosed air stream of 50 liters per minute. The porosity of the mask material shall be tested as specified in 4.5.1.1 and (b) the filtering efficiency shall be not less than 95 percent when tested as specified in 4.5.1.2.

3.1.1.2 Nosepiece. The nosepiece shall be fabricated from a single strip of malleable aluminum, rounded on both ends. The dimensions shall be 0.019 $\pm$ 0.002 inch thick, 3-1/2 $\pm$ 1/64 inches long and 3/16 $\pm$ 1/64 inch wide; shall be tested as specified in 4.5. The strips shall be malleable enough to be readily bent to, and maintain, the contours of the wearer's nose, so as to prevent the loss of expired air between the mask and the face in this area.

3.1.1.2.1 Color. The color of the nosepiece shall be the natural color of the metal, blue-green, or white.

3.1.1.3 Elastic band. The elastic band that is attached to the sides of the mask shall be fabricated from heat-resistant natural rubber or synthetic latex rubber. The length of the unstretched elastic band between attachment points shall be 15 inches $\pm$ 1 inch. The elastic band shall extend at least 3/8 inch beyond each staple. The width of the elastic band shall be not less than 1/16 inch nor more than 3/16 inch and shall be not less than .020 inch nor more than .030 inch thick. The elastic band shall not deteriorate when subjected to autoclaving and shall be tested as specified in 4.5.1.4

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3.1.1.3.1 Color. The color of the elastic band may be white, blue, or blue-green to match the body of the mask.

3.1.1.4 Staples. The staples used to secure the elastic band to the sides of the body of the mask shall be fabricated from flat, galvanized, low-carbon steel wire. The stapling wire shall have a thickness of 0.020 inch ± 0.003 inch.

3.1.1.4.1 Color. The staples shall be the natural color of the metal used.

3.1.2 Type II mask.

3.1.2.1 Facing material. The facing material shall be nonwoven fabric of 100 percent viscose rayon fiber. The material shall be colorless, or white, or blue, or green and shall be odorless.

3.1.2.2 Filter media. The filter media shall consist of a filter mat composed of pure flame-blown or finespun glass fibers, or polypropylene fibers. The glass fibers shall have an average maximum diameter of 1.5 microns and the polypropylene fibers shall have an average diameter of 4 microns. The filter media shall not contain any spicules of glass and shall be odorless. The filter mat may have a light green or blue innocuous color added.

3.1.2.3 Tie tapes. The tie tapes shall be made of woven cotton material or constructed from nonwoven fabric. The tie tapes shall be constructed by sewing or shall be impregnated with a thermoplastic, water-resistant adhesive. The tie strings shall finish 3/16 to 7/16 inch wide, and shall not show fraying or loose threads along the cut edges.

3.1.2.3.1 Color. The color of the tape shall be white, or dark blue or dyed to match the color of the mask.

3.1.2.4 Nosepiece. The nosepiece shall be fabricated from a single strip of malleable aluminum, rounded on both ends. The strip shall be 3-1/4 to 4-1/2 inches long and 1/8 to 1/4 inch wide. The thickness shall be 0.019 ± 0.002 inch. When the nosepiece is to be placed on inside of the mask, it may be fabricated as described above or from flat, malleable, aluminum strips measuring 0.125 ± 0.005 inch wide and 0.020 ± 0.002 inch thick, and extending the length of the mask. The strip shall be malleable enough to be readily bent, and to maintain the contours of the wearer's nose, so as to prevent leakage at the top of the mask and fogging of spectacles.

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3.1.2.4.1 Color. The color of the nosepiece shall be blue, blue-green, or white. When the nosepiece is placed inside the mask it may be the natural color of the metal.

3.2 Construction.

3.2.1 Type I mask. The body of the mask shall be molded or formed into a cup-shape, the periphery of which shall be contoured to fit over the bridge of the nose, under the eyes, along the cheeks, and under the chin. The cupping of the body shall be sufficiently deep so that the wearer's nose and mouth do not touch the mask. The portion of the cup over the nose and mouth shall be ribbed or corrugated transversely. The metal nosepiece shall be permanently cemented to the outside surface of the mask and approximately centered over the nose area with the ends not more than 3/8 inch from closest edge of mask. Each end of the elastic band shall be permanently attached to the outside surface of the mask with a metal staple. Each staple shall be attached 1/8 to 5/8 inch from edge of mask but not in the rib area. The elastic band and staple shall be assembled so that the elastic band does not break or pull out, and the staples do not pull out of the body of the mask when tested as specified in 4.5.1.3. The length of the unstretched elastic band, when measured between the points of attachment shall be 15 inches $\pm$ 1 inch. The body of the mask shall serve as a filter. The characteristic of the formed coated fabric shall be such that the body of the mask shall retain its shape under normal handling and use, but will permit adjustment and conformance to the contours of the face. The body of the mask shall not have an offensive odor.

3.2.1.1 Autoclaving of complete mask. The complete mask shall be capable of being sterilized in an autoclave without any deleterious effect, however a discoloration of the staples during autoclaving may occur. Complete mask shall be tested as specified in 4.5.1.4.

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3.2.2 Type II mask. The mask shall consist of a filter mat encased in two soft layers of nonwoven fabric. The filter mat shall be constructed of the material specified in 3.1.2.2. The layers of nonwoven fabric shall be constructed of the material specified in 3.1.2.1. The mask shall have a pouch or it shall be pleated to allow for a snug fit at the edges and without binding at the nose or mouth. When the mask has a pouch, the pouch shall be formed by a deep fold across the mask. The sides of the pouch shall be heat sealed. The peripherial margins of the masks shall be bound with the tape specified in 3.1.2.3. The tape along the horizontal or vertical margins shall be extended to serve as tie tapes. The four tie tapes shall be a minimum of 13 inches long from each side of the mask. When the tapes are tied the mask must conform to the contour of the wearer's nose and cheek. A nosepiece as specified in 3.1.2.4 shall be placed on the outside or inside the top horizontal edge of the mask, as applicable. When the nosepiece is on the outside of the mask, it shall be secured so that it will not detach during use or when soaked with perspiration. When the mask is constructed with the metal strip inside the top horizontal edge, the word "INSIDE" shall be legibly stamped in ink on the middle of the top horizontal inside margin, except when the cover material on the inside of the mask is of a different color than the cover material on the outside.

3.2.2.1 Dimensions. The finished mask shall have the following dimensions:

Horizontal (Length)------6-3/8 to 7-3/4 inches.
Vertical (Width)---------3-3/4 inches $\pm$ 1/2 inch.
Pouch, when applicable:
  Depth----------1-1/2 inches $\pm$ 1/8 inch.
  Width (at narrowest point)-----4-1/2 inches $\pm$ 1/8 inch.
  Pouch starts------1-1/2 inches $\pm$ 1/8 inch from top.
  Snout angle-------33° to 37° from vertical margin.
  Maximum width of heat seal--------1/4 inch.
Pleats, when applicable:
  Opened width-------6-1/2 inches to 7-3/4 inches.
Tie Tapes------A minimum of 13 inches long from each side of the mask.

3.2.2.2 Performance characteristics.

3.2.2.2.1 Filtering efficiency. The filtering efficiency of the mask shall be not less than 96 percent when tested by any of the methods of 4.5.2.1, except 4.5.2.1.2 (Method-II). When tested by 4.5.2.1.2 (Method-II), the minimum filtering efficiency of the mask shall be 90 percent.

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3.2.2.2.2 Air exchange pressure. (Ease of breathing). The air exchange pressure of the mask shall be a maximum of 5 millimeters (.197 inch) of H_2O, per square centimeter of sample tested. The air exchange pressure of the mask shall be tested as specified in 4.5.2.2.

3.3 Workmanship. The mask shall be free of cuts, tears or any other imperfections or defects that will detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of conformance. Where certificate of conformance is submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be tested in accordance with all the requirements of referenced specification, and standard unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105, except when otherwise specified herein.

4.4 Examination of the end item. The masks shall be examined for visual defects. The sample unit for this examination shall be one mask. The inspection level shall be S-4 of MIL-STD-105. The acceptable quality levels shall be 0.65 for major defects and 2.5 for minor defects, expressed in defects per hundred units.

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4.4.1 Classification of defects. Examination shall be conducted in accordance with, but not restricted to the following classification of defects.

4.4.1.1 Type I mask.

Categories and defects

Major

101	Body of mask not molded or formed into a cup-shape.
102	Body of mask not contoured to face.
103	Ribs omitted or do not transverse the body of the mask.
104	Any holes, cuts or tears in body of mask.
105	Nosepiece omitted or not securely attached.
106	Nosepiece not malleable to the extent specified.
107	Elastic band omitted or not securely attached.
108	Mask having offensive odor.

Minor

201	Visible spots or specks of foreign matter on body of mask.
202	Wrinkles on body of mask.
203	Staples not placed as specified.
204	Nosepiece not placed as specified.
205	Edge of mask not trimmed.

4.4.1.2 Type II mask.

Categories and defects

Major

101	Any hole, cut or tears in body of mask.
102	Mask without pleats or pouch.
103	Nosepiece omitted.
104	Nosepiece not malleable to the extent specified.
105	Nosepiece not securely attached to outside of mask, when applicable.
106	Mask constructed without four tie strings.
107	Mask not bound on all four sides.

Minor

201	Visible spots or specks of foreign matter on body of mask.
202	Mask not clean (not free of lint or loose fibers).
203	The word "inside" not marked on inside top horizontal margin, when specified.
204	Tie strings having thread ends not trimmed at the end of stitching operation, when applicable.

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4.5 Tests. Tests shall be conducted to determine compliance with specification requirements, and as indicated in tables I and II. The unit of product for testing shall be one mask. The sample size shall be determined in accordance with special inspection level S-2 of MIL-STD-105, except for the characteristic "filtering efficiency" which shall be determined in accordance with special inspection level S-1. The number of test failures shall be zero.

Table I. - Tests (Type I mask)

	Characteristic	Requirement Paragraph	Test Method
Fabric	Identification of polyester and rayon	3.1.1.1	1/
	Fabric construction	3.1.1.1	Visual
	Color	3.1.1.1.1	Visual
	Porosity	3.1.1.1.2	4.5.1.1
	Filtering efficiency	3.1.1.1.2	4.5.1.2
Nosepiece	Metal; identification of aluminum	3.1.1.2	1/
	Dimensions	3.1.1.2	Visual
	Color	3.1.1.2.1	Visual
Elastic band	Material Identification	3.1.1.3	1/
	Width	3.1.1.3	Visual
	Thickness	3.1.1.3	Visual (Gauge)
	Autoclaving	3.1.1.3	4.5.1.4
	Color	3.1.1.3.1	Visual
Mask	Elastic & Staple assembly	3.2.1	4.5.1.3
	Length of elastic band between staples	3.2.1	Visual
	Autoclaving	3.2.1.1	4.5.1.4
Staples	Metal, Identification Low-carbon galvanized steel wire	3.1.1.4	1/
	Thickness, gauge	3.1.1.4	Visual
	Color	3.1.1.4.1	Visual

1/ A certificate of conformace shall be submitted by the supplier and will be acceptable for the stated requirement.

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4.5.1 Type I mask.

4.5.1.1 Porosity test. The porosity of the mask material shall be tested across a circular sample, encompassing the ribbed part of the mask, having an area of 17.5 square inches. This area shall be exposed, perpendicular to the air flow, in a closed air stream having a continuous flow rate of 50 liters of air per minute. The pressure differential from the inside of the mask to the outside, shall be measured on a standard inclined manometer.

4.5.1.2 Filtering efficiency. The filtering efficiency of the mask shall be determined by the method of Greene and Vesley (J. Bacterial Vol. 83, No. 3, P-663, March 1962), or a comparable "in vitro" method using the Andersen Sampler (see 4.5.3).

4.5.1.3 Elastic and staple assembly. The elastic and staple assembly shall be tested by grasping the elastic band midway between the staples and extending it twice to a distance of 23 inches from the staples. The elastic shall be completely relaxed between each extension.

4.5.1.4 Autoclaving. The complete mask shall be subjected to autoclaving at 250° Fahrenheit, 15 pounds pressure for not less than 20 minutes.

Table II. - Tests (Type II mask)

	Characteristic	Requirement Paragraph	Test Method
Facing material	Identification of material (Rayon)	3.1.2.1	1/
	Fabric construction	3.1.2.1	Visual
	Color	3.1.2.1	Visual
Filter media	Identification of filter media	3.1.2.2	1/
	Fiber diameter (Average)	3.1.2.2	1/
Tie Tapes	Identification of material	3.1.2.3	1/
	Color	3.1.2.3.1	Visual
Nosepiece	Identification of material (metal)	3.1.2.4	1/
	Dimensions	3.1.2.4	Visual
	Thickness	3.1.2.4	Visual (gauge)
	Color	3.1.2.4.1	Visual

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Table II. - Tests (Type II mask) (Con't)

	Characteristic	Requirement Paragraph	Test Method
Mask	Width of tie tapes	3.1.2.3	Visual
	Dimensions	3.2.2.1	Visual
	Filtering efficiency	3.2.2.2.1	4.5.2.1
	Air exchange pressure	3.2.2.2.2	4.5.2.2

1/ A certificate of conformance shall be submitted by the supplier and will be acceptable for the stated requirement.

4.5.2 Type II mask.

4.5.2.1 Filtering efficiency. The filtering efficiency of the type II mask shall be determined in accordance with 4.5.2.1.1 or 4.5.2.1.2 or by method of Greene and Vesley (J. Bacterial. Vol. 83, No. 3, P-663, March 1962), or a comparable "in vitro" method using the Andersen Sampler (see 4.5.3).

4.5.2.1.1 Method-I. Cultures---Staphylococcus aureus, Utah strain 15, isolated from a clean surgical wound infection is used for all bacterial filtration efficiency tests. Cultures are transferred at least 3 times in nutrient broth, Baltimore Biological Laboratory (BBL) before preparation of the test suspension. Each transfer is incubated at 37 degrees Centigrade for 24 hours. Cultures are constantly agitated throughout the incubation period on a mechanical shaking machine to disperse the organisms as much as possible.

Aerosol filtration of bacteria. The efficiency of each mask is tested with the Andersen sampler (see figure 1), a device which discriminates aerosol particles by size and enumerates viable particles in the aerosol on nutrient agar, BBL, and Petri plates.

A standard procedure is followed for all tests. The 24 hour culture of Staphylococcus aureus is adjusted to a standard optical density of 75 as measured in a Klett-Summerson photoelectric colorimeter with a blue filter in the light path. The adjusted culture is then diluted 1:100,000 in a sterile, nutrient broth.

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Para. 4.5.2.1.1 Continued

The diluted culture suspension is fed into a nebulizer from a syringe driven by a constant drive mechanism (see figure 2). The nebulizer is operated at 15 pounds per square inch pressure to produce an aerosol in an aerosol chamber. An aerosol sample of Staphylococcus aureus is drawn through at a flow rate of one cubic foot per minute for one minute, without interposing a filter material. The nebulizer is then turned off and the chamber washed with 1 cubic foot of air drawn through the chamber and sampler. The plates are removed from the sampler and replaced immediately with fresh dishes. Thereafter, each filter mask to be tested is interposed, in turn, between the aerosol chamber and the sampler, and a comparable aerosolized sample of Staphylococci are drawn through the apparatus. All plates are incubated at 37 degrees Centigrade for 48 hours and the colonies counted. Every precaution is taken to maintain all factors constant, except for the different filtration masks. The air flow is measured by Roger Gilmont Instruments (RGI) flowmeter or a flowmeter of comparable precision. The filtration efficiency is reported in percent efficiency:

$$\text{Percent efficiency} = \frac{C-F}{C} \times 100$$

Where C = number of colonies on control plates.

F = number of colonies on plates with filter masks interposed.

4.5.2.1.2 Method-II. The apparatus used in method II is shown in figure 3. An aerosol of bacteria is produced by a combination of air pressure at points "I" and "A" and vacuum at points F, G, and H. The flow rates are 1,900 milliliters per minute at the positive pressure end and 3,320 milliliters per minute at the vacuum end of the apparatus. The positive pressure is directed through a Vaponephrin nebulizer (point B) containing approximately two milliliters of a suspension of 5,000 viable cells of Staphylococcus aureus 209 per milliliter of 0.1 percent peptone water. This suspension is pipetted into the nebulizer. Such a suspension is prepared from frozen viable cells which shall be counted prior to test.

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Para. 4.5.2.1.2 Continued

The aerosol is directed into a mixing chamber (point C) in which the larger droplets of the bacteria laden aerosol are removed. The aerosol is drawn by a controlled vacuum applied at the vacuum end, through the face mask secured in the mask holder (point D) and are trapped on the millipore filter membrane which is 47 mm in diameter having a porosity of 0.8 u (micron) and placed at Point E. Alternate samples are taken with and without masks in the system and a time period of five minutes is required for each sample tested. Each filter membrane is placed onto a nutrient agar plate and incubated for twenty-four hours at 37° C. At the end of the incubation period the colonies shall be counted using a dissecting microscope or a Quebec Colony Counter. The filtering efficiency of the masks is calculated as follows:

$$\text{Efficiency in \%} = \frac{\text{(Count without mask)-(Count with mask)}}{\text{count without mask}} \times 100$$

4.5.2.2 Air exchange pressure. A device which measures the pressure differential required to draw air through a measured surface area at a constant air flow rate is used to measure the air exchange pressure of various materials (see figure 4). Water filled manometers, (m_1 and m_2) are used to measure the pressure differential. A Roger Gilmont Instrument (R.G.I.) flowmeter or a flowmeter of comparable precision is used for measurement of the air flow. An electric vacuum pump draws air through the apparatus and a needle value is used to adjust the flow rate. The test material is placed across the 2.5 cm diameter orifice and clamped into place so that the test material will be in line and across the flow of air. The pump is started and the flow of air adjusted to 8 liters per minute. Manometers m_1 & m_2 are read and recorded. This procedure is carried out on five different areas of the mask and the readings averaged. The air exchange pressure is calculated as follows:

$$\Delta P = \frac{\bar{X}_{m_2} - \bar{X}_{m_1}}{4.9}$$

Where: $\bar{X}\, m_1$ = mm water pressure, Manometer No. 1 average of 5 test areas, low pressure side of test material.

$\bar{X}\, m_2$ = mm water pressure, Manometer No. 2-average of 5 test areas, high pressure side of test material.

4.9 = cm^2 area of test material.

ΔP = Pressure differential per cm of test material expressed as mm of water.

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NOTE: In lieu of two manometers, a Differential Inclined Draft Gauge, graduated in tenths and calibrated in inches of water may be used. Pressure differential may be reported in inches of water.

4.5.3 When alternate comparable tests, not listed in this specification, are selected to determine filtering efficiency, the test procedure shall be reviewed for approval by the Directorate of Medical Materiel, ATTN: ATT, of the Defense Personnel Support Center, Philadelphia, Pa. 19101.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One package (50 masks), as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in the following table shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table III. Packaging Quantities

Type	Unit package	Intermediate package	Exterior container
I	1 unit	12 units	48 units
II	1 unit	6 units	24 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be over-packed in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package.

5.1.3.1.1 Type I mask. Unit package shall be a carton of appropriate size constructed in accordance with PPP-B-566, except that commercial colors shall be acceptable. Closure shall be adequate to prevent spilling of contents under normal handling.

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5.1.3.1.2 Type II Mask. Unit package shall be a carton of appropriate size and design. Carton shall be so constructed as to permit individual (one-at-a-time) dispensing of masks. Closure of carton shall be adequate to prevent spilling of contents under normal handling.

5.1.3.2 Intermediate package.

5.1.3.2.1 Type I Mask. Intermediate package shall be a carton of appropriate size constructed in accordance with PPP-B-566, except that commercial colors shall be acceptable. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.3.2.2 Type II Mask. Intermediate package shall be a carton of appropriate size constructed in accordance with PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.2 Level B.

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5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The masks covered by this specification are intended for use by Military Medical Services.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type required (see 1.2).
(c) Selection of applicable levels of packing. (see 5.1 and 5.2).

6.3 This specification does not include all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types which are normally procured to meet Military requirements.

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Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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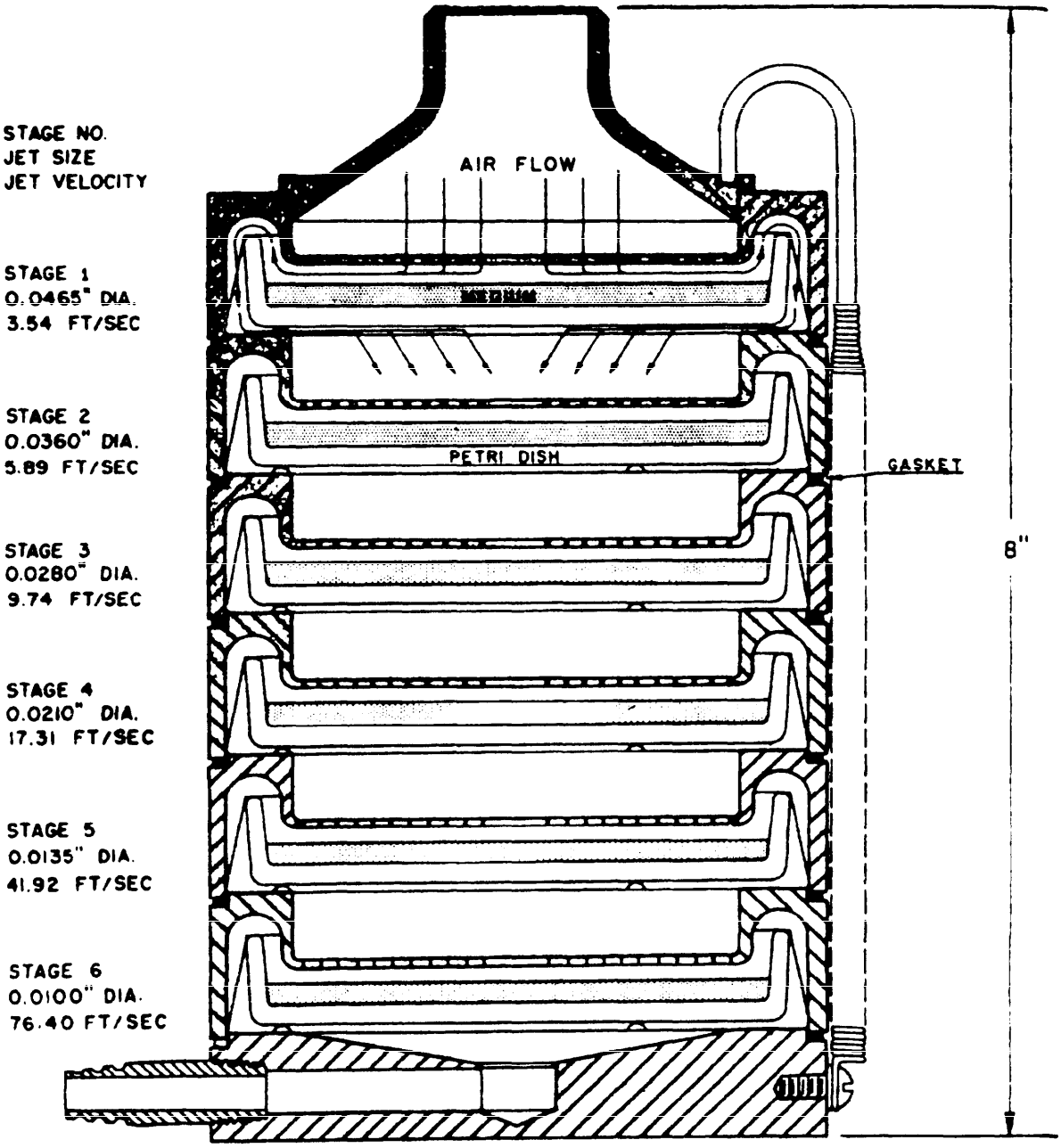


Figure 1

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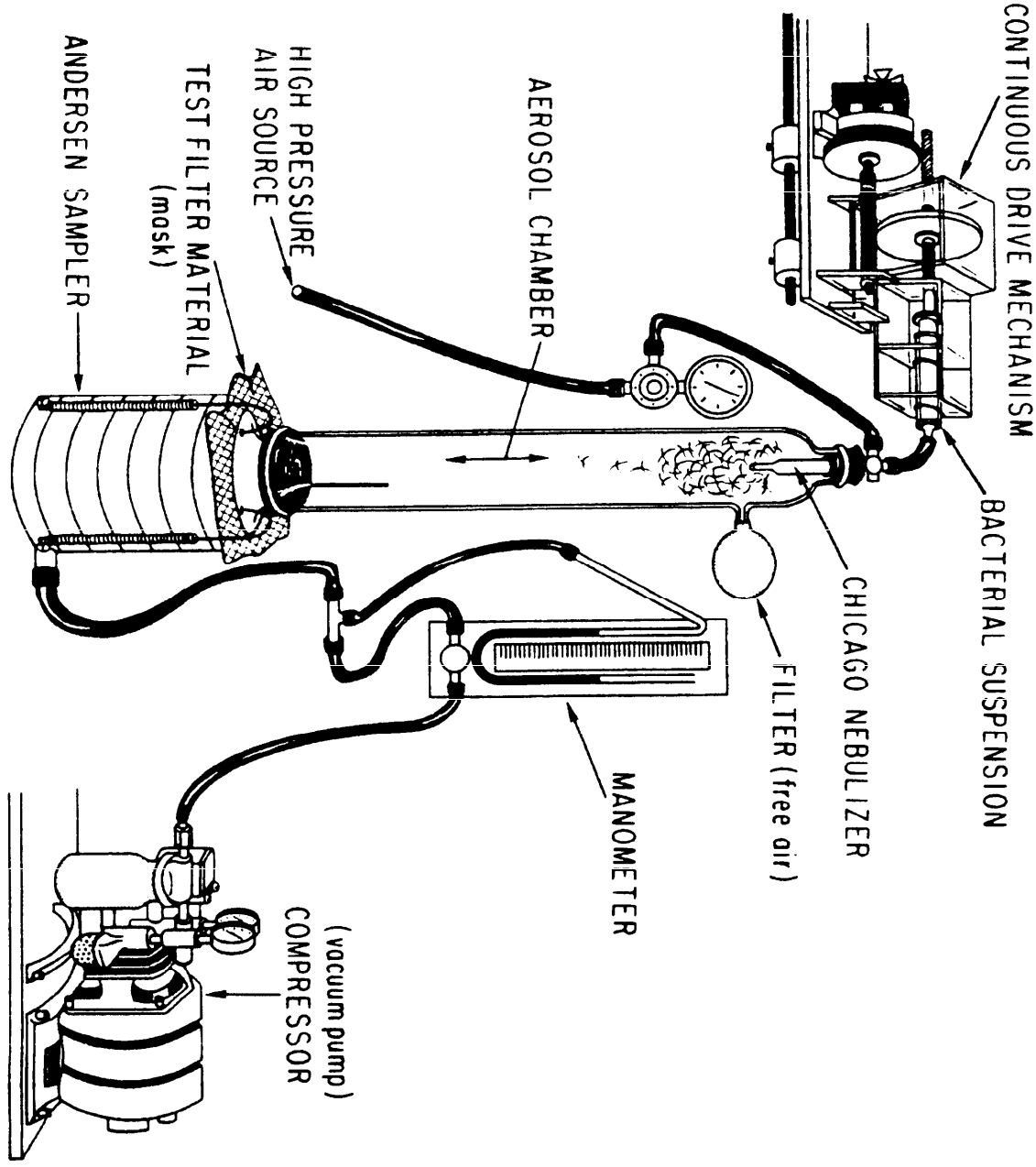
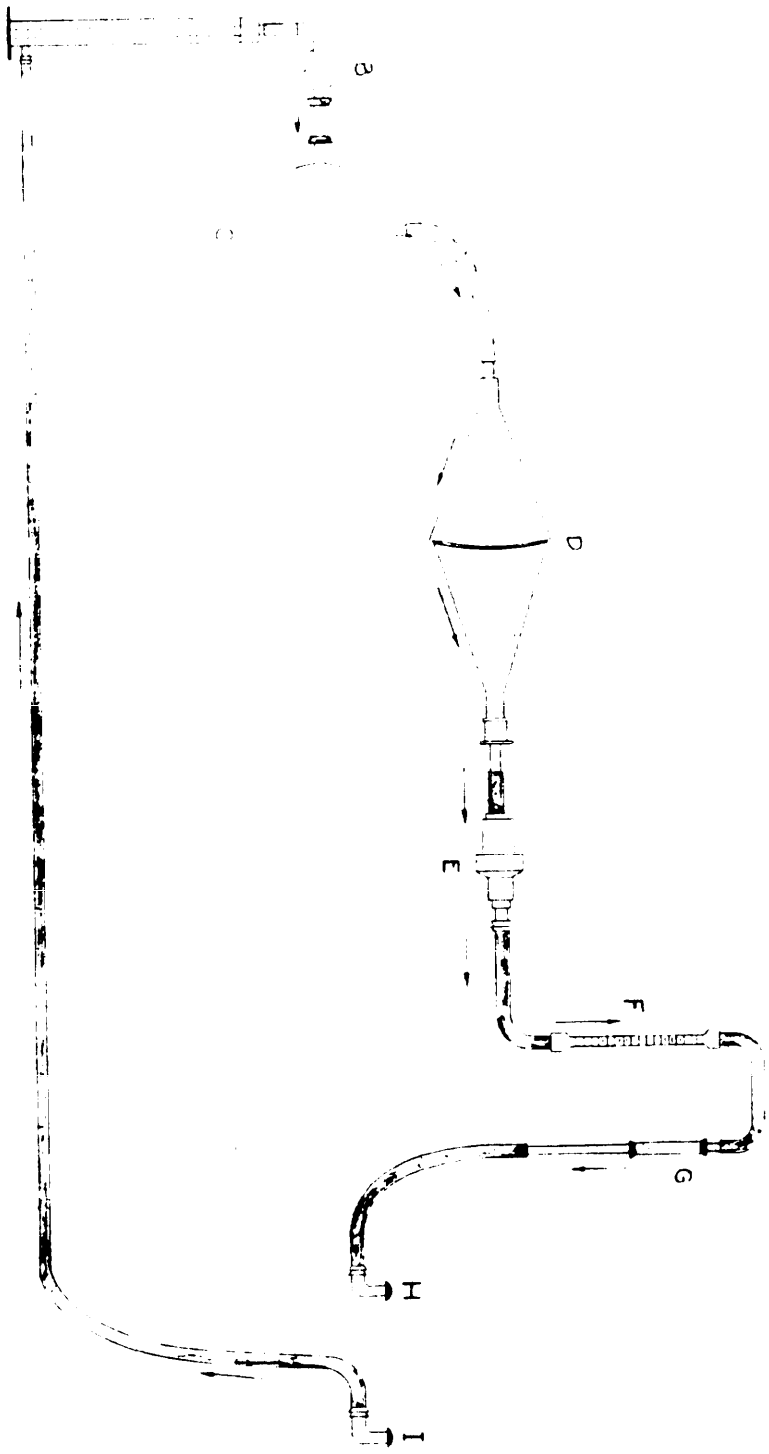


Figure 2

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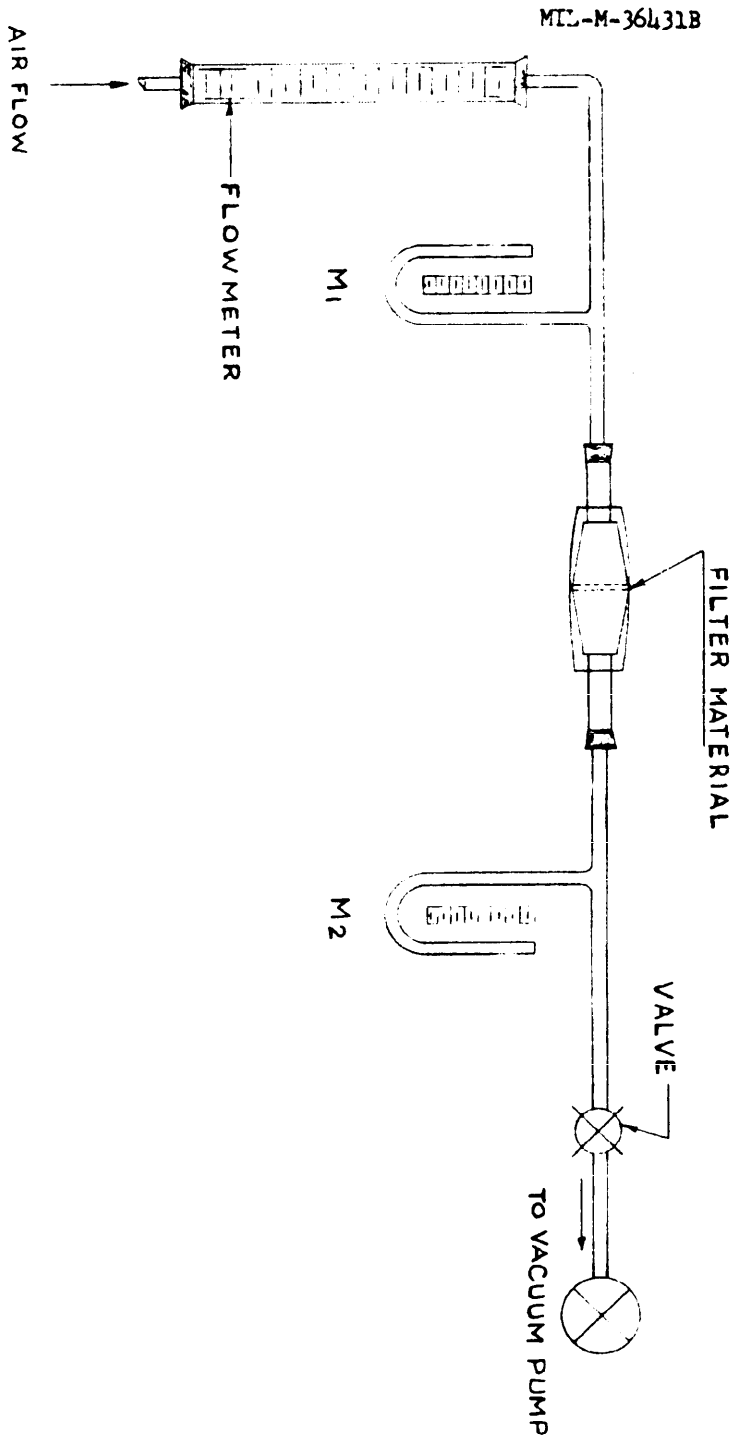


A POSITIVE AIR FLOWMETER
B NEBULIZER
C GLASS MIXING CHAMBER
D MASK HOLDER
E MILLIPORE FILTER & HOLDER
F VACUUM FLOWMETER
G SETTING CHAMBER
H VACUUM
I AIR

BACTERIAL AEROSAL FILTRATION APPARATUS

FIGURE 3.

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AIR PRESSURE-FLOW MEASURING APPARATUS

Figure 4

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements.

SPECIFICATION

MIL-M-36431B FSN

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

Fold

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DPSC FORM 2418
MAR 68

EDITION AUG 66, WILL BE USED UNTIL EXHAUSTED

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